D.4.4. TOOL SELECTION

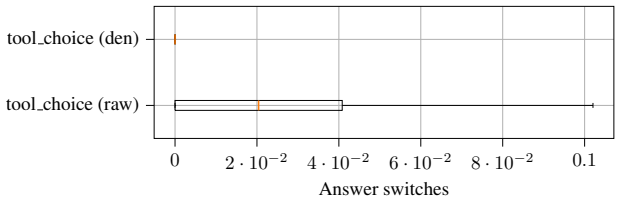


Figure 51. Distribution of the mean of answer switches for raw and denoised outputs in Qwen3-30B-A3B, evaluated on the Tool Selection.

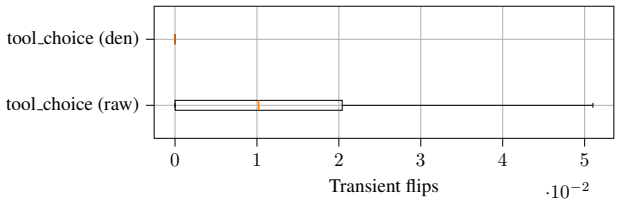


Figure 52. Distribution of the mean of transient flips for raw and denoised outputs in Qwen3-30B-A3B, evaluated on the Tool Selection.



Figure 53. Distribution of the mean of tokens after final answer switch for raw and denoised outputs in Qwen3-30B-A3B, evaluated on the Tool Selection.

E. Training Details

Positive and negative training examples. Probe training examples are constructed by sampling intermediate reasoning steps from each trace rather than using all steps. Positive examples correspond to reasoning states at which computation could be halted without changing the final answer, whereas negative examples correspond to states where additional reasoning is still required.

For each example, we sample a fixed number S of reasoning steps from the trace. For a sampled step i, we extract the hidden state of the final generated token at that step from all transformer layers, denoted h_i^l for layer l. Steps satisfying $i > t^\star$ are assigned a positive label, while steps satisfying $i \leq t^\star$ are assigned a negative label. For examples with $t^\star = -1$, all sampled steps are treated as positive. The value of S is treated as a training hyperparameter.

Because many examples exhibit no answer switch (i.e., $t^\star = -1$), positive examples are more frequent than negative examples. To mitigate this imbalance, we oversample negative examples so that the two classes are approximately balanced within each minibatch.

Optimization. We train the probe using the Adam optimizer. Learning rates are selected from the candidate set $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 2 \times 10^{-3}, 5 \times 10^{-3}, 1 \times 10^{-2}, 2 \times 10^{-2}, 5 \times 10^{-2}, 1 \times 10^{-1}\}$. After fixing the best learning rate, we select the probe layer by choosing the layer whose probe achieves the highest AP on validation. The probe parameters from the best (learning rate, layer) configuration are used for all reported results for a particular model and dataset.

F. Experiments on other datasets

We present metrics as defined in Section 3.3 on more datasets in Table 25

Table 25. Summary of answer trajectory metrics on additional datasets.

Dataset	Accuracy (%)	$\Pr(t^\star = -1)$ Raw% / Denoised%	AnswerSwitches Raw / Denoised	TAFs(T)	T_{after} Raw / Denoised
Humanity's Last Exam	5.2	39.2 / 48.9	10.8 / 1.3	6.5	1186 (49.9%) / 1765 (67.1%)
GPQA Diamond	56.1	35.5 / 43.8	10.9 / 1.5	6.6	1608 (45.7%) / 2259 (59.8%)
AIME26	72.7	27.2 / 27.2	14.3 / 2.1	7.1	660 (16.4%) / 1654 (32.9%)